Back-Projected Attention Fields and Foveated Reconstruction: Phenomenology and Computation of Retro-Indexed Gaze in Recorded Media

Thomas Oury

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Abstract

Recording folds time by linking a subject's performance at capture (t_0) to audience detection at replay (t_2). We propose two complementary, interface-first frameworks: (1) a **Back-Projected Attention Field (BPAF)**, a phenomenological–computational model describing how subjects modulate performance in response to an inferred future audience embodied by the camera; and (2) a **Foveated Reconstruction** method that estimates a low-dimensional lighting/saliency structure from aggregate eye-tracking traces without access to the original footage. We treat "retro" language as a time-indexing of constraints on a measurable interface, not an ontological commitment. Distinct metaphysical interpretations are treated as gauge choices unless the environment forces a distinction into accessible observables. The result is a unified, testable picture: the felt presence of future viewers is an operational variable at t_0, and the structure that guides viewing at t_2 can be partially reconstructed (up to an equivalence class) by inversion of gaze data.

1 Setup: Time-Folding as an Interface

Cameras detect photons; displays emit photons; audiences detect again. In a forward index, this is capture $\rightarrow$ archive $\rightarrow$ replay. In a retro-index, constraints associated with replay are *back-projected* to shape present control at capture.

We adopt an *interface-first* constraint: we speak only in terms of accessible observables and realized interventions. We do not pay for unobserved universes.

Timeline

- t_0: capture / performance.
- t_1: storage / edit / distribution pipeline.
- t_2: replay / detection by viewers.

2 Methodological Constraint: No Counterfactual Ontology

Definition 1 (Observable Interface)**.** *Let $\mathcal{I}$ denote the interface consisting of (i) observables R and (ii) realized interventions Σ available to the experimenter. A claim is admissible iff it is stated as a mapping over $\mathcal{I}$.*

Figure 1: Time-Folding as Interface

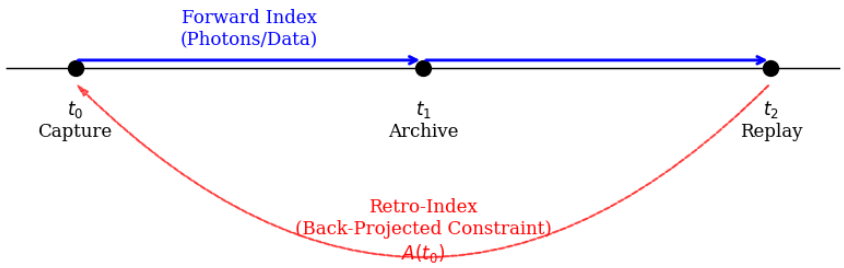

Figure 1: Spacetime diagram showing forward index versus retro-index constraint arrows. The forward path follows capture → archive → replay, while retro-indexed constraints flow backward from t_2 to shape control at t_0.

Definition 2 (Interface Equivalence (Gauge Equivalence))**.** *Two models/frames M_1, M_2 are* equivalent on *$\mathcal{I}$ if they induce the same distribution over observables for every realized intervention:*

$$\forall \sigma \in \Sigma : \quad \Pr(R \,|\, \sigma, M_1) = \Pr(R \,|\, \sigma, M_2).$$

If $\mathcal{I}$ does not force a distinction, we treat the difference as representational gauge.

This paper uses "retro" language as *indexing* (constraints at t_2 shaping behavior at t_0 through the subject's model and media pipeline), while remaining agnostic about deeper metaphysics unless forced by $\mathcal{I}$.

3 Back-Projected Attention Field (BPAF)

Subjects often report feeling "watched" by a camera even in an empty room. We model this as a back-projected constraint field: an operational quantity that modulates performance control at t_0.

Definition 3 (Audience Distribution and Stakes)**.** *Let $\mathcal{V}$ be a distribution over viewer-classes v as represented by the subject (explicitly or implicitly), and let $w(v) \geq 0$ encode stakes/weight (relevance, status, expected audience size, personal meaning).*

Definition 4 (Back-Projected Attention Field)**.** *Define the BPAF at capture time t_0 as*

$$A(t_0) \;=\; \mathbb{E}_{v\sim\mathcal{V}}\Big[w(v)\, K(t_2 - t_0)\Big], \tag{1}$$

where $K(\Delta t)$ is a temporal kernel encoding how replay-horizon influences present control (bounded/decaying/context-dependent).

Definition 5 (Performer State Update)**.** *Let $s(t)$ denote a performer-control state (prosody, timing, gesture policy, expressive gain). We posit an update rule*

$$\Delta s(t_0) = \Psi(A(t_0),\, c(t_0),\, \theta)\,, \tag{2}$$

where $c(t_0)$ are camera/studio cues (lens salience, tally light, operator ritual) and θ captures stable traits ("amplifiers" vs "inhibitors").

Figure 2: The BPAF Control Loop

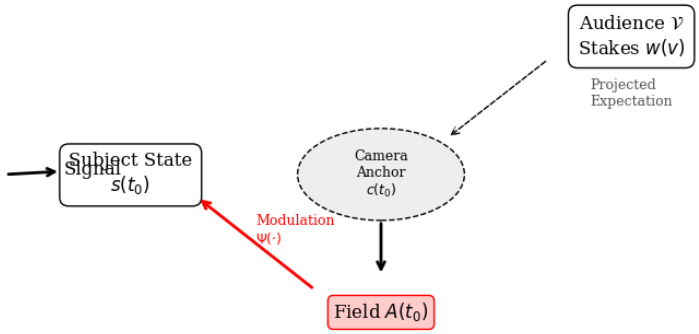

Figure 2: BPAF schematic showing the flow from audience distribution $\mathcal{V}$ and stakes $w(v)$, through camera cues $c(t_0)$, to the attention field $A(t_0)$, and finally to performer state updates $s(t_0)$. The operator Ψ mediates this transformation based on performer traits θ.

3.1 Measurable Readouts

Operationalize $s(t_0)$ via multimodal features:

- prosody: pitch range, intensity, pause structure;
- kinetics: gesture rate, beat-gesture synchrony;
- face: expressivity entropy / action-unit dynamics;
- text: emphasis markers, surprisal shifts;
- human judgments: blinded rater discrimination.

3.2 Phenomenological Predictions (as Operators)

Proposition 1 (BPAF Modulation Under Stakes)**.** *Holding camera cues $c(t_0)$ fixed, increases in represented stakes (in $w(v)$) induce systematic shifts in $s(t_0)$, with direction and gain controlled by θ.*

Proposition 2 (Blinded Discrimination)**.** *If BPAF-driven shifts are large enough relative to natural variance, blinded raters can discriminate high-stakes vs low-stakes capture conditions above chance using only performance signals.*

These predictions are compatible with social facilitation effects (perceived presence modulates arousal/performance) while keeping the paper interface-first [1].

4 Gauge Anchoring: Why the Camera Compresses the "Audience Problem"

Definition 6 (Gauge Anchor (Performer-Side))**.** *A* gauge anchor *is a solver/performer-maintained representation that stabilizes the semantics of observation and control across time. In this context, the camera acts as an anchor for "who is watching," enabling stable performance policies across takes.*

Lemma 1 (Gauge-Anchored Compression of Audience State)**.** *Let $\mathcal{V}$ be the subject's represented audience distribution. Increasing camera salience and ritual cues $c(t_0)$ reduces representational drift by anchoring $\mathcal{V}$ to a canonical frame ("the lens as the viewer proxy"). This compresses the effective audience-state space, stabilizes the meaning of stakes $w(v)$, and increases operator reliability in $\Psi(\cdot)$ across takes.*

Corollary 1 (Expert Performance as a Ratchet)**.** *Skilled performers exhibit a stable anchor (low representational drift) and a reusable operator library (repeatable control policies) that preserve achieved communicative constraints while improving subsequent takes.*

5 Foveated Reconstruction: Inverting Gaze to Recover Structure

At replay, each viewer produces gaze signals that partially reveal stimulus structure plus biases.

Definition 7 (Gaze Signal)**.** *Let $g_i(x, y, t)$ denote viewer i's gaze density (or fixation point process) over image coordinates (x, y) and time t. Let $\bar{g}$ denote an aggregate across viewers (e.g., mean density).*

5.1 Forward Model (Saliency-to-Gaze)

We posit a forward operator from latent structure to gaze:

$$\bar{g}(x, y, t) \;\approx\; \Phi\Big(L(x, y, t);\; P\Big), \tag{3}$$

where $L(x, y, t)$ is a latent lighting/saliency field, Φ is a perceptual operator, and P encodes stable biases (faces, motion, center bias, task). Classical and modern saliency models motivate Φ [2, 3].

5.2 Inverse Problem (What Is Recoverable?)

We estimate L consistent with observed gaze:

$$L^{=\arg\min_L \;\|\Phi(L;P)-\bar{g}\|^2+\lambda R(L),}(4)$$

where $R(L)$ is a regularizer (smoothness, natural-scene priors, piecewise lighting zones).

Lemma 2 (Non-Identifiability and Equivalence Class Recovery)**.** *In general, Φ is not injective: many L map to similar $\bar{g}$. Therefore L should be interpreted as an* equivalence-class representative *under Φ and P. The reconstruction target is not the original footage but an invariant summary: a "lighting/saliency contour map" (key/fill zones, highlight topology, shadow structure, saliency gradients) sufficient for downstream inference.*

5.3 Ratcheting the Inversion

The inversion improves when we add stable constraints (a "ratchet") using additional interface structure:

- segment by shot type (CU/MS/WS), face-on vs profile;
- incorporate priors (face region during speech, motion energy);
- multi-audience aggregation reduces idiosyncratic noise;
- edit-boundaries and cut detection constrain $L(x, y, t)$.

Corollary 2 (Tell $\neq$ Footage; Tell $\Rightarrow$ Structure Under Ratchet)**.** *A gaze heatmap is not a key to footage pixels, but under repeated constraints and regularization it becomes a key to stable structural summaries of the stimulus.*

Figure 3: Gaze Inversion & Structural Recovery

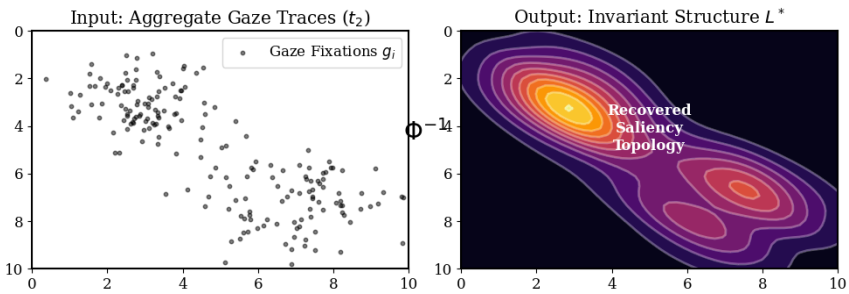

Figure 3: Gaze inversion workflow showing the transformation from aggregate gaze $\bar{g}$ through the perceptual operator Φ combined with regularization, to recover the latent lighting/saliency structure L^* as a contour map. The inversion operates on equivalence classes rather than exact pixel recovery.

6 Regimes as Geometry: (G, B, T) for Recorded Media

To match our broader paper style, we introduce a low-dimensional regime description.

Definition 8 (Grounding Strength G)**.** *Let $G \in [0, 1]$ denote the degree to which external constraints dominate internal patching. In this paper, "external constraints" include both (i) camera-side cueing that stabilizes the audience proxy at t_0, and (ii) replay-side empirical constraint from aggregate gaze at t_2.*

Definition 9 (Integration Bandwidth B)**.** *Let $B \geq 0$ denote the effective integration capacity (how well a system integrates cues, feedback, and internal state into coherent control updates). At t_0 this is performer integration; at t_2 this is model/inversion integration.*

Definition 10 (Memory Horizon T)**.** *Let $T \geq 0$ denote the temporal window over which state summaries and setpoints remain stable and usable for control (take-to-take continuity; shot-to-shot continuity).*

R1: Anchored Performance (High G, Sufficient B, T)

Stable camera-as-proxy, repeatable performance operators, monotonic improvement across takes.

R2: Stage-Fragility (High A, Low B)

High perceived stakes without integration capacity yields constriction, jitter, non-monotonic takes.

R3: Free-Running Performance (Low G)

Weak anchoring; audience model drifts; performance becomes internally consistent but poorly tuned to downstream detection.

R4: Reconstruction Success (High G on replay, Sufficient B, T)

Aggregate gaze constrains the inverse; L converges to a stable equivalence-class representative.

7 Experiments (Interface-First)

7.1 Phenomenology Protocol (Realized 2×2)

- **Stakes cue:** low vs high (instructional framing + distribution plan).
- **Camera proxy cues:** minimal vs salient (lens prominence, tally light, operator ritual).

Outcomes: multimodal performance features + blinded rater discrimination + self-report of "felt audience."

Figure 4: Media Regimes (G, B, T)

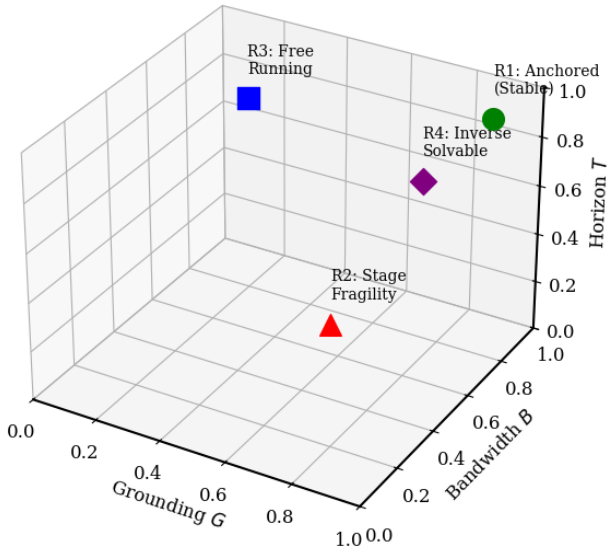

Figure 4: Regime map in the (G, B, T) parameter space showing labeled regions R1–R4. Grounding strength G represents external constraint dominance, integration bandwidth B represents system integration capacity, and memory horizon T represents temporal stability. Each region corresponds to distinct behavioral and computational regimes described in the text.

7.2 Reconstruction Protocol

Collect aggregate gaze on footage with known lighting (production notes or controlled studio):

- fit Φ (or use established predictors) and solve for L;
- evaluate recovery of *structure* (zones/topology), not pixels;
- test ratcheting improvements via segmentation and multi-viewer aggregation.

8 Implications

- **Psychology:** the "felt presence" of future viewers is an operational control variable at capture [1].
- **Media studies:** recordings are time-folding interfaces linking emitter control at t_0 to detector traces at t_2.
- **Computation:** gaze inversion recovers invariant stimulus structure up to an equivalence class [2, 3].
- **Style:** the camera operates as a gauge anchor that collapses representational drift and enables a ratchet.

9 Conclusion

We presented an interface-first account of retro-indexed gaze in recorded media. BPAF formalizes how subjects at t_0 control performance under a back-projected audience constraint. Foveated reconstruction formalizes how aggregate gaze at t_2 can be inverted to recover an invariant lighting/saliency summary without original footage. We treat "retro" as a representational choice unless the environment forces a distinction

into observables. Either way, the operational result is the same: recordings are structured couplings across time.

References

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